

Lab 2: Data Publishing

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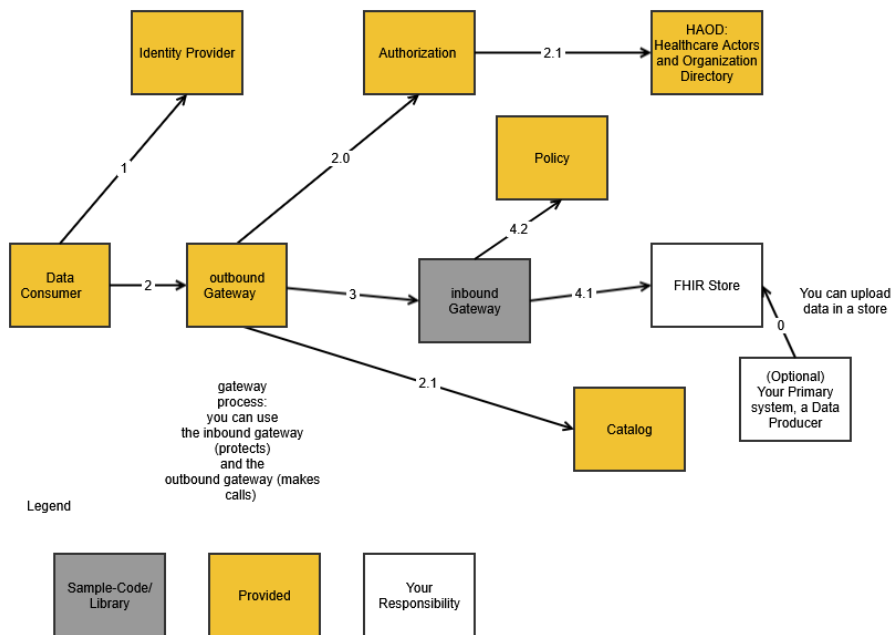
Context

We are working in the context of SwissHDS. The SwissHDS reference architecture describes the conceptual model.

Primary system vendors can publish their own data in the HDS. There are 3 options how to do this:

- a) Your primary system copies data into a shared FHIR Data Store (e.g. hosted by an intermediary). This is the simplest solution.
- b) Your primary system exposes data directly via a FHIR API (following HDS rules for aspects such as access control).
- c) Your primary system copies data into your own dedicated FHIR Data Store. Most complex solution.

We have set up a simplified SwissHDS reference environment. **It is not designed for real person data — only for test data. Also credentials such as tokens used must not be able to access production data. The interfaces and UIs are simplified.**



Goals

- Patient mock data from your primary system is published in the SwissHDS in a standard FHIR format.
- Other players in the SwissHDS can securely access the data (if they have network visibility of your gateway!)

Steps

1) Translating primary system data into FHIR data product instances

We suggest using a Medication Plan or eTOC data product as defined at <https://fhir.ch/ig/ch-emed/index.html> or <https://fhir.ch/ig/ch-etoc/index.html> (we expect to refine the existing standards for the SwissHDS, but this will be based on the current standards). These data product types are expected to gain increasing regulatory relevance in the near future (through the HMG/Therapeutic Products Act and for patients moving among institutions).

We assume that some primary system vendors already support this format. The translation can happen either **at publication time** (when copying into a separate Data Store — option a)) or **only at access time**.

We provide example resources you can download in the demo UI.

The identifier of the corresponding patient is stored in the Patient resource of each document bundle. It can be found e.g. under `resource.identifier.0.value`

2) Exposing data behind a standardized FHIR interface

As introduced above, this can be implemented via options **a), b) or c)**.

Common steps for all options:

- Publish your Data Store (or more precisely the URL of the gateway that secures it) in the **Data Product Catalog**. In case you use a store of us (option a)), we have already done this for you.
Refer to [Data Product Catalog Usage](#) for info on how to use it.
- Add the provider identity=the url of your gateway both in your gateway and the System tab of the Policy service, refer to [Policy Service Usage](#).
- Ensure the Data Product Instances can be accessed:
 - Patients can directly access their own Data Product Instances (matched via the patient's Id (HDS Id)).
 - A practitioner P can access a Data Product Instance if
 - The practitioner P is a practitioner in the system (=it has a GLN number in the HAOD)
 - The practitioner P has an active relationship with the patient of the Data Product Instance (the patient's Id is again the HDS Id).
The active relationships are managed in the [Policy service](#).
 - The Data Product Instance is not in the deny rules of the Policy service.
 - Refer also to [Lab 1: Login and data consumption#Howelementsarelinkedtoeachother](#) or our sample data.

2a) Using an existing separate Data Store instance

(For now, we have provisioned a simple store of the kind an intermediary might provide. SwissHDS does not prescribe the API between the primary system and the intermediary.)

The store is available at: <https://swishds.u.c.bag.admin.ch/spital-a/ui/> It is a regular HAPI FHIR Store. You can use its API to upload data products.

To simulate the upload, you can use the unsecured FHIR UI (or API) of our Data Store (behind the gateway). THIS IS JUST A SIMULATION AND WILL BE DIFFERENT LATER!

1. Upload the data product via the UI or by using a standard FHIR POST command.

2b) Exposing data products via an existing native FHIR interface of the primary system

The exact steps depend on the capabilities of your primary system. See variants in §7.3.1 (C) b. Here we focus on variant (C) b.i. with a HDS Gateway. In other words, we assume that a FHIR endpoint already exists, from which the data product can be retrieved.

Sub-steps:

1. For the FHIR endpoint to behave in compliance with SwissHDS, it must follow several rules, for example:
 - It must validate access rights via the Policy Service, including filtering returned data products accordingly.
 - It must log all accesses in the Audit Log (we defer this for later).
 - It needs some IGs (FHIR Implementation Guides), such as the IGs of the served Data Products and generic IGs for Bundle Searches/ RefChain/ Custom SearchParameter
We have provided these IGs in a zip.
Refer to section [HowwequerydataproducintheFHIRStore](#) for details.
2. We have written a sample implementation for this: the **HDS Gateway SwissHDS Gateway Usage**. We recommend using this gateway here, or at least studying its source code to understand how to connect your system.
We use the inbound or inbound gateway for this operation: Other parties call the gateway and the gateway forwards requests to your FHIR store.
3. An important consideration is securing the connection between the HDS Gateway and the existing FHIR endpoint. We do not prescribe how this is done. One could assume that the protection is done on the network level (i.e. that one does not expose the underlying Data Store API).
We exclude this aspect for the moment.
4. Remark: If you want to use our UIs or services, the URL of your gateway must be reachable from our sandbox (51.103.226.251.), and all other participants that want to share data with.

2c) The primary system copies data into your own dedicated FHIR Data Store

Setup the gateway as in step 2b) and setup your own dedicated FHIR Data Store (e.g. HAPI FHIR).

Copy your data from your primary system to the data store as in step 2a).

3) Verifying that the access works

We have implemented a test page for this purpose. It performs a HDS login and accesses a data product. We assume that practitioner XY (authenticated via Keycloak) wishes to access the product.

Sub-steps:

1. You can use an existing practitioner account from Keycloak or create a new one. Here is the list of demo practitioners: [Environment-Details for Labs#Anmeldedaten](#)
2. The practitioner must be registered in the HAOD. Refer to the details here: [HAOD Usage](#)
3. The practitioner must have the rights to access the Data Products of the patient.
A practitioner P can access a Data Product Instance if
 - The practitioner P is a practitioner in the system (=it has a GLN number in the HAOD)
 - The practitioner P has an active relationship with the patient of the Data Product Instance (the patient's Id is again the HDS Id). The active relationships are managed in the [Policy service](#).
 - The Data Product Instance is not in the deny rules of the Policy service.
4. Access the data product via our Demo Client: <https://swisshds.u.c.bag.admin.ch/client/login>
5. Alternatively, a primary system that was connected to SwissHDS in Lab 1 could also access the data product.

Troubleshooting hints

- Refer to the hints of lab 1: [Lab 1: Login and data consumption#Troubleshootinghints](#)
- Do you have the search IG for FHIR added to your data store? (If you use option b) or c)?)

Infrastructure and Support

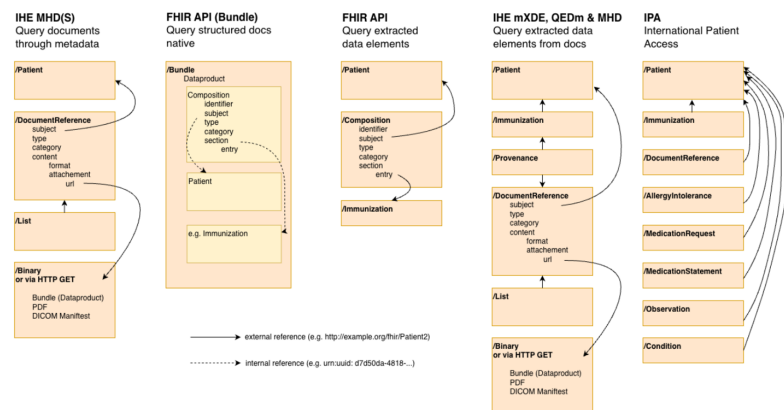
Key documents are located in the GitHub project <https://github.com/SwissHDS/swiss-hds-labs> and are sent to you by email. If anything is unclear, Issues can be created in the same Github repository.

Remark: How we query data product instances in the FHIR Store.

There are different ways to query data product instances. We currently use the second variant from the left (with a FHIR API that queries bundles). Normally, your FHIR store needs a special IG to support the needed query parameters.

FHIR API for querying data products

The FHIR API defines a RESTful API and standard search parameters. The following picture illustrates the different possibilities how a data product can be queried through the FHIR API endpoints:



Questions

1. How much effort did it take you to finish the lab?
2. What IdP(s) did you use? Why?
3. What variant did you use a), b) or both?
4. Did you use the provided HDS Gateway? If yes, what were experiences?
5. Screenshots/ Film über Integration
6. Other remarks?